

Electrolytic cells

Learning outcomes

By the end of this section you should be able to:

- Explain how current is conducted in an electrolytic cell.
- Deduce the products of the electrolysis of a molten salt.

You have surely held an aluminium can in your hand once or twice. Aluminium cans are used daily to hold a wide range of products, from carbonated soft drinks to canned fruits and vegetables, soups, aerosol sprays and even some pharmaceutical products.



Credit: Thomas-Soellner, Getty Images



Credit: Aleksandr Durnov, Getty Images

Figure 1. A variety of products are kept in aluminium cans.

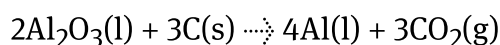
Have you ever stopped to consider where all the aluminium for aluminium cans comes from?

Electrolytic cells

Many elements are not found in nature in their elemental form. They are found as ores, where they occur as ions bonded with other ions (see [subtopic S2.1](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43108/>)). In order to change from the ion to the atom, there must be electron transfer (see [subtopic R3.1](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43484/>)).

Aluminium, as well as other chemicals needed by humans, can be produced by using electricity in electrolytic cells. You may recall that electrolytic cells use electrical energy to produce chemical compounds (see [section R3.2.5](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electrodes-in-electrochemical-cell-id-46489/>)). The process uses electrical energy to drive a non-spontaneous chemical reaction and is called electrolysis.

In the previous example, electrolytic cells can be used to extract aluminium metal from its ores by reducing the aluminium ions in molten aluminium oxide, Al_2O_3 (also known as alumina), using graphite as electrodes. The reaction to produce pure aluminium metal and carbon dioxide gas according to the reaction:



This is a non-spontaneous reaction which means energy needs to be provided for the reaction to occur. That energy comes from the use of electricity in the form of a battery or a power generator.

All electrolytic cells consist of:

- a single container
- two inert electrodes (generally graphite is used)
- a solution (the electrolyte)
- a battery which can be considered an electron pump (DC power source).

The first step in creating an electrolytic cell is to melt the salt that contains the ions that will be reduced and oxidised. Remember that solid ionic salts are bad conductors of electricity (see [subtopic S2.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43108/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43108/>)), which means they cannot be electrolysed.

Making connections

Under what conditions can ionic compounds act as electrolytes? (see [subtopic S2.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43108/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43108/>)).

Ionic compounds will act as electrolytes when subjected to high temperatures and an electric current so that they become molten. In the liquid state, the lattice structure is disrupted and the cations and anions are free to move and will be able to reach the electrodes to undergo either oxidation or reduction, respectively.

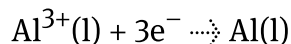
Ionic compounds will also be able to act as electrolytes when they are dissolved in a polar solvent, such as water, and dissociate into their constituent ions. In a polar solvent, the positive and negative poles of the solvent molecules interact with the positively and negatively charged ions, causing them to separate and become surrounded by solvent molecules in a process called solvation or hydration.

In the example of aluminium oxide, Al_2O_3 , the salt needs to be exposed to a high temperature and an electric current to undergo electrolysis. The melting point of aluminium oxide is 2000°C however, cryolite is added to lower the melting point. Cryolite is sodium aluminium fluoride (Na_3AlF_6), a colourless to white halide mineral. Its addition to the electrolyte helps to reduce the cost of the process, as much less energy is required to heat the ore to a temperature of around $960\text{--}980^\circ\text{C}$, the temperature at which the aluminium oxide melts, allowing the ions to move freely in the liquid. The electric current applied to the electrodes inserted into the molten aluminium oxide then separates the aluminium and oxygen ions as each species travels to one of the electrodes.

The electrodes used in this type of cell are inert, as oxidation and reduction should occur only with the ions produced by the molten salt. Platinum (Pt) is a commonly used material for electrodes. In a school lab setting, generally, graphite is used as it is a cheaper option and a good conductor of electricity (see [subtopic S2.2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43109/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43109/>)) but relatively inert, which allows the electrical current to pass to the electrolyte without interfering with the reactions happening at each electrode.

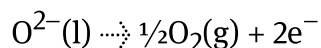
The battery supplies electrons to the cathode, where reduction occurs. From the ions available in the electrolyte, Al^{3+} and O^{2-} , the most likely to receive electrons and become reduced is the aluminium ion, which leads to the following half-reaction:

Cathode(reduction):



Because the cathode is gaining electrons from the power source, it has a more negative charge than the anode. This makes the anion in the electrolyte, O^{2-} , travel to the anode to lose electrons and become oxidised, according to the half-reaction:

Anode(oxidation):



Notice that the product here is oxygen gas and will be visible as bubbles being produced at the anode. The oxygen that is produced reacts with the carbon of the positive electrodes, forming carbon dioxide, so they gradually burn away, which adds to the cost of the process. As a result, the positive electrodes have to be replaced frequently.

Watch the video below about the electrolysis of alumina.

How To Extract Aluminium By Electrolysis | Environmental C...

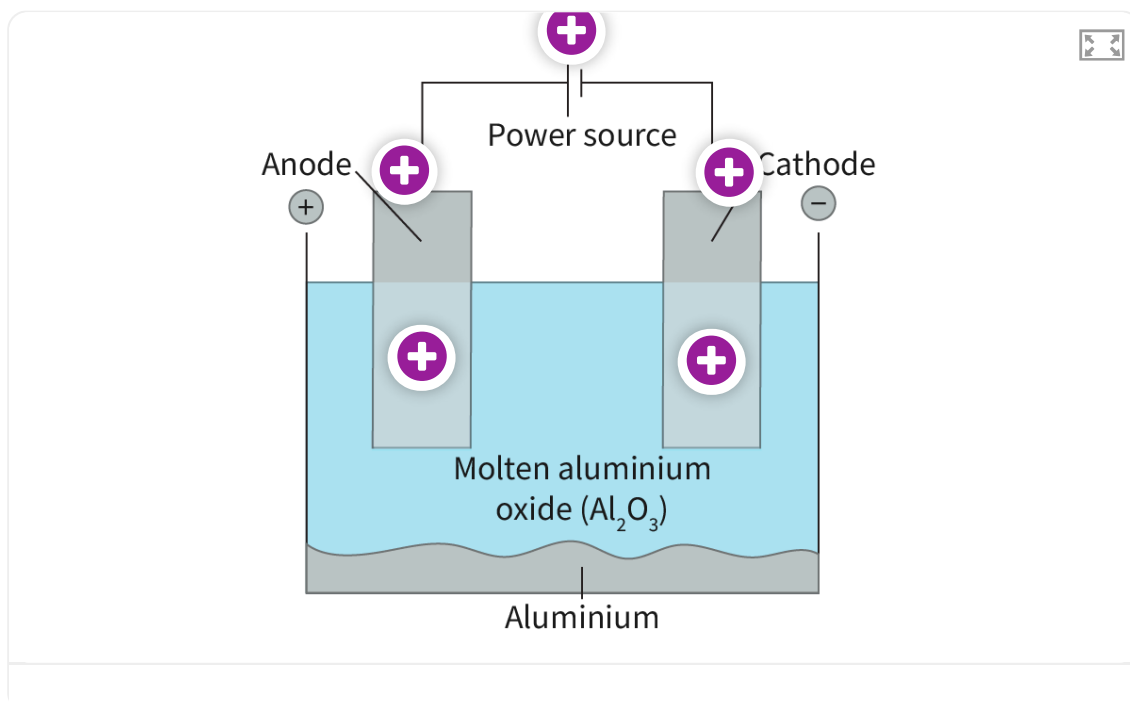


Video 1. The electrolysis of aluminium oxide.

At the cathode, liquid aluminium will be collected and can then be separated from the electrolyte due to its density difference.

In the external wire connecting both electrodes, there is a current of electrons moving from the anode to the cathode, just like in the voltaic cells, but the electrodes have reversed potentials (the anode is positive and the cathode is negative). The circuit is closed by the movement of the ions in the electrolyte (cations to the cathode and anions to the anode).

Use **Interactive 1** to investigate what is happening at each location in the electrolysis of aluminium oxide.

**Interactive 1.** Electrolysis of Molten Aluminium Oxide.

 More information for interactive 1

An interactive diagram displays the electrolysis of molten aluminum oxide (Al_2O_3). Two vertical electrodes are immersed in the molten aluminum oxide. The left electrode is labeled anode and has a positive sign, while the right electrode is labeled cathode and has a negative sign. A power source is connected to both electrodes. The bottom of the container is labeled aluminum, representing the deposited metal.

There are 5 Hotspots represented by plus signs at various places. Hotspot 1 is next to the label anode. Hotspot 2 is above the power source. Hotspot 3 is next to the label cathode. Hotspot 4 on the anode. Hotspot 5 on the cathode. Clicking on the hotspots reveals what is happening in each of the places.

The following items are revealed at respective hotspots:

Hotspot 1: It displays electron flow exiting from the anode and entering the positive terminal of the power supply.

Hotspot 2: It displays the electron flow from the negative terminal to the positive terminal.

Hotspot 3: It displays the electron flow from the negative terminal entering the cathode.

Hotspot 4: It displays an O^{2-} moving towards the anode and liberating two electrons and an O_2 molecule released from the electrolyte.

Hotspot 5: It displays 3 electrons moving down the cathode and an Al^{3+} ion moving towards the cathode inside the electrolyte and releasing Al ion from the cathode.

Study skills

Remember that the construction of electrolytic cells should include a DC power source connected to anode and cathode and an electrolyte.

Look back at **Figure 2** in section R3.2.5

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electrodes-in-electrochemical-cell-id-46489/>) for all the details required on a diagram of an electrolytic cell.

To work out what happens during the electrolysis of molten salt, follow these steps:

1. Identify all species present (the anions and cations that can be released when the salt melts).
2. Identify which species are attracted to the anode (positive electrode) and which species are attracted to the cathode (negative electrode).
3. Deduce the two half-reactions, knowing the oxidation happens at the anode and reduction happens at the cathode.
4. Determine the direction for the flow of electrons in the external wire and of the ions in the electrolyte.

5. State what will be observed at each electrode based on the products of the two half-reactions.

Worked example 1

Consider the electrolysis of a molten salt such as lead(II) bromide.

State the ions that will be present in the electrolyte.

Lead(II) bromide, PbBr_2 , when molten, will release the cation Pb^{2+} and the anion Br^- into the electrolyte.

Worked example 2

Consider the electrolysis of a molten salt such as lead(II) bromide.

State the half-reaction happening at the anode and deduce the product formed and what you will be able to observe.

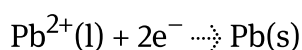


At the anode, oxidation will occur. Two anions of Br^- will move to the anode and will lose two electrons to become bromine gas, $\text{Br}_2(\text{g})$. A brown gas will be visible, bubbling around the anode.

Worked example 3

Consider the electrolysis of a molten salt such as lead(II) bromide.

State half-reaction happening at the cathode and deduce the product formed and what you will be able to observe.



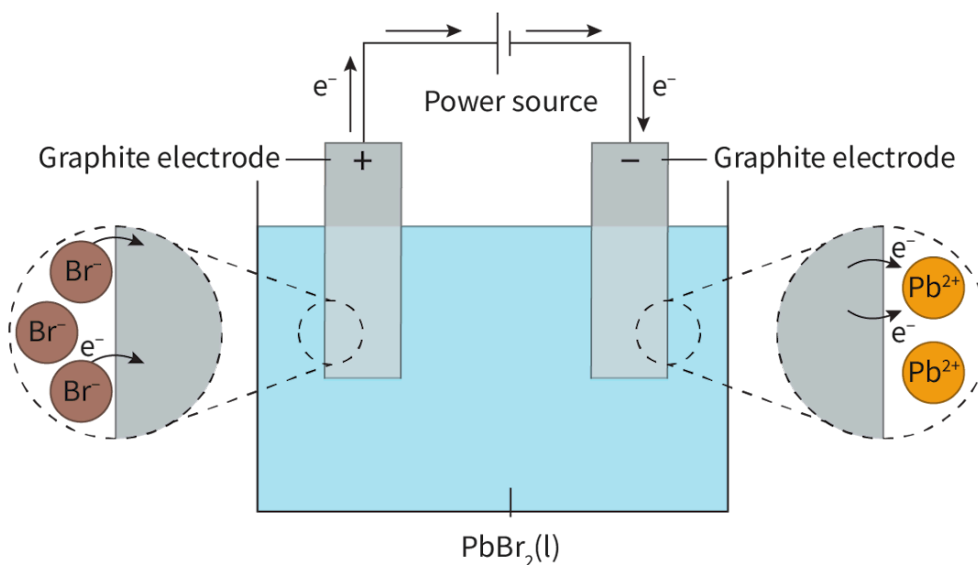
At the cathode, reduction will occur. Each cation Pb^{2+} will move to the cathode and will gain two electrons to become lead metal, Pb(l) . The lead will be visible around the cathode.

Worked example 4

Consider the electrolysis of a molten salt such as lead(II) bromide.

Draw and annotate a diagram of the electrolytic cell, including the movement of the ions and electrons, as well as the charges for each electrode.

The diagram should look similar to this one.



Electrolysis of molten lead(II) bromide.



Worked example 5

Consider the electrolysis of a molten salt such as lead(II) bromide.

Discuss why lead(II) bromide molten conducts electricity while it does not when in a solid form.

Molten lead(II) bromide has free ions that can move. Electricity is passed when there are charged particles able to move freely. In the solid form, all the ions that make up lead(II) bromide are fixed into a lattice structure and, therefore cannot freely move. This means they cannot conduct electricity.

Aspect: Experiments

The concept of electrolysis and its uses were developed by several scientists over time, but the modern understanding of the process can be traced back to the work of Michael Faraday in the early 19th century. Faraday's experiments on electrolysis showed that an electric current could cause chemical reactions to occur in a solution or molten material.

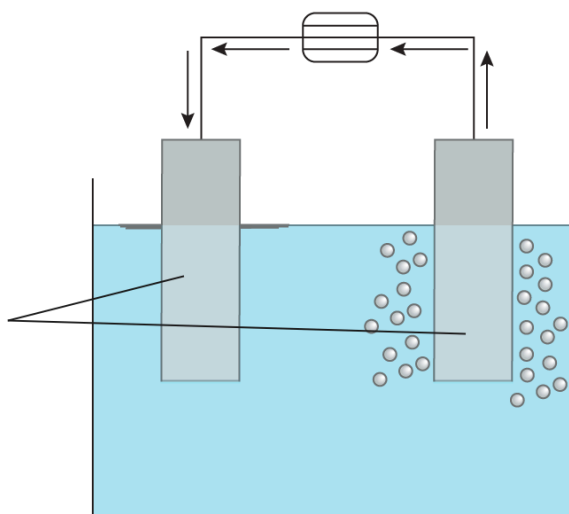
Nowadays, electrolysis is used in various industries and applications, like the production of aluminium from alumina discussed at the beginning of this section. Other reactions include:

- Electrolysis of water, used to produce hydrogen gas for fuel cells and other applications.
- Electroplating, used to deposit a thin layer of metal onto a surface using an electric current; it improves the appearance, corrosion resistance, or wear resistance of various metal objects, such as jewellery, car parts, and electronics.
- Electrolytic refining, used to purify metals such as copper, nickel, and zinc, which are used in various industries, such as electronics and telecommunications.

Activity

- **IB learner profile attribute:** Inquirer
- **Approaches to learning:** Communication skills — Using terminology, symbols, and communication conventions consistently and correctly
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual/pair

The electrolysis of calcium chloride (CaCl_2) is used on a small scale to produce pure calcium metal for laboratory use. Drag all the labels to the correct spaces on the image that illustrates the electrolysis of calcium chloride.



Molten calcium
chloride CaCl_2

Anode

Graphite

Battery

Cathode

-

+

$\text{Cl}_2(\text{g})$

$\text{Ca}(\text{l})$

e^-

e^-

✓ Check

Interactive 2. Drag the labels to the correct spaces.